

Appendix B: Cross-Method Robustness and Concordance of Scale-Adaptive Interaction Test Results via GAMM and Aligned Rank Transform (ART)

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Introduction

The primary goal of this vignette is to assess the **robustness and concordance** of scale-dependent entropic index $\times$ group interaction discoveries from Scale-Adaptive Interaction Tests (SAIT): do they generalize to a non-parametric statistical framework with fewer distributional assumptions? Agreement between GAMM and ART establishes robustness under complementary modeling assumptions.

Two Complementary Approaches

Method 1: Paired regression-spline mixed model (GAMM-style) with functional q-curve structure (full details in the main vignette). TSENAT historically refers to this family as GAMM; it is fitted with `nlme::lme`, not `mgcv::gamm()`.

- Framework: Semi-parametric model $Y = f_1(q) + f_2(\text{condition}) + f_3(q, \text{condition}) + \text{subject-intercept} + \varepsilon$ with smooth rather than linear f terms.
- Paired fits: `nlme::lme` with `ns(q, df = 3)  $\times$  condition`, subject random intercepts, and a continuous-q AR(1) working correlation within subject $\times$ condition (`nlme::corCAR1`, $\exp(-\phi|q_i - q_j|)$; validated by Monte Carlo simulation); the interaction is tested with a marginal F-test.
- q is a deterministic functional argument (not time): the confirmatory hypothesis is the functional interaction $H_0 : \beta(q) = 0$ for all q, tested on the original H(q) curve. Gaussian residuals are assumed (paired mixed-model paths do not support extended families); heteroscedasticity is detected but variance weights are not applied in paired designs.

Method 2: Aligned Rank Transform (ART) — rank-based non-parametric interaction testing (ARTool (Kay et al. 2021)), with Hochberg step-up FWER correction. The Conover-Iman Rank Transform is available via `method='rt'`.

- ART strips main effects before ranking (“alignment”) to preserve interaction structure (Higgins and Tashtoush 1994; Wobbrock et al. 2011). It makes no Gaussian-error assumption, but is not assumption-free: validity depends on the design, exchangeability, alignment and sample size.
- q is treated as a categorical factor (its continuous ordering and the AR(1) working correlation across q are not modeled) — the fundamental difference from GAMM.
- Monte Carlo caveat: unpaired ART is mildly anti-conservative at small n (Conover-Iman `rt` recommended there). A full treatment follows in the “Aligned Rank Transform (ART)” section below.

Why Two Methods for One Question?

Statistical testing in transcriptomics faces a fundamental challenge: no single method is universally optimal. Different approaches trade-off power for robustness:

Aspect	Semi-Parametric (GAMM)	Non-Parametric (ART)
Assumptions	Normality, homoscedasticity	No Gaussian errors; validity still requires a valid design, exchangeability and alignment
Power	Highest (if assumptions hold)	Good; reduced but robust to violations
Interaction testing	Marginal F-test on regression-spline interaction terms	Proper interaction tests via alignment (Higgins and Tashtoush 1994)
q-value treatment	Continuous smooth predictor (preserves ordinal structure)	Categorical factor (discards ordering)
Robustness	Moderate	Highest
Outlier sensitivity	Moderate potential for bias	Minimal; inherently resistant

Robustness strategy: Concordance between both methods provides evidence of robustness across statistical frameworks (it does not by itself prove either method correct).

Setup

This section initializes the analysis environment by loading required packages, setting a random seed for reproducibility, and preparing the test dataset.

```
suppressPackageStartupMessages({
  library(ggplot2)
  library(SummarizedExperiment)
  library(dplyr)
  library(gridExtra)
})

set.seed(42)

if (!requireNamespace("devtools", quietly = TRUE)) {
  stop("Please install devtools to run this vignette against the local package source.")
}
if ("package:TSENAT" %in% search()) {
  detach("package:TSENAT", unload = TRUE, character.only = TRUE)
}

pkg_root <- normalizePath("../", mustWork = FALSE)
devtools::load_all(pkg_root)

# Load preprocessed dataset
data(readcounts)
readcounts <- as.matrix(readcounts)
```

```

mode(readcounts) <- "numeric"

# Load metadata
metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)

gff3_dataset <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

# Configure analysis parameters first (best practice: fail-fast validation)
config <- TSENAT_config(
  sample_col = "sample",
  condition_col = "condition",
  subject_col = "paired_samples",
  q = seq(0, 2, by = 0.05),
  nthreads = 3,
  paired = TRUE,
  control = "normal"
)

# Build analysis object: creates SummarizedExperiment and initializes TSENATAnalysis
analysis <- build_analysis(
  readcounts = readcounts,
  tx2gene = gff3_dataset,
  metadata = metadata_df,
  tpm = tpm,
  effective_length = effective_length,
  config = config
)

# Roles: tpm -> abundance QC in filter_analysis(); effective_length ->
# length-normalized entropy from RAW counts. The two are mutually exclusive
# in a single computation (TPM already incorporates length normalization).

# Apply filtering for quality control
analysis <- filter_analysis(
  analysis,
  stringency = "medium"
)

```

Computing Tsallis Entropy with Pseudocount Regularization

Pseudocounts are an **optional regularization mechanism**. RNA-seq experiments typically contain many zero or near-zero counts, which creates two practical problems for rank-based methods:

1. **Ties in rankings:** Sparse counts produce many tied values (especially zeros), which reduces the discriminatory power of rank tests. Rank-based statistics depend on unique orderings; when many observations are identical, the test statistic contains less information, reducing statistical power.
2. **Library size artifacts:** Small count differences due to sequencing depth variations can overshadow biological differences. Normalized library sizes (accounting for sequencing depth) can make entropy estimates more comparable across samples (Robinson et al. 2010; Love et al. 2014).

By adding small pseudocounts we achieve two benefits:

- **Regularization** breaks ties and reduces zero-inflation effects on rank-based inference.
- **Comparability** makes entropy estimates more stable across samples with different sequencing depths.

Importantly, a pseudocount is a regularization *choice*, not a standard RNA-seq normalization procedure, and it **changes the estimand**: for

$$p_i = \frac{x_i + a}{\sum_j (x_j + a)}$$

you are no longer measuring the empirical transcript composition. This is especially consequential for low-abundance genes and $q < 1$ (where small probabilities are amplified) and for $q = 0$ (where a positive pseudocount changes the support).

The `calculate_diversity()` function implements automatic pseudocount selection, scaling pseudocount magnitude and diversity estimates by median library size to ensure robust statistical inference across the range of sequencing depths in your dataset.

```
# Compute diversity using S4 wrapper with bootstrap confidence intervals
analysis <- calculate_diversity(
  analysis,
  norm = TRUE,
  pseudocount = "auto"
)
```

Aligned Rank Transform (ART)

TSENAT uses the **Aligned Rank Transform (ART)** as the default non-parametric method for testing Q×Condition interactions. ART is a well-established rank-based non-parametric approach for factorial analysis, implemented via the ARTool R package (Kay et al. 2021; Wobbrock et al. 2011). The classical Conover-Iman Rank Transform is also available via `method='rt'` as an alternative approach (Conover and Iman 1981).

1. Robustness to Distribution Violations

Tsallis entropy values exhibit inherent distributional properties that often violate parametric assumptions:

- Non-normality: Entropy commonly exhibits bounded support (e.g., normalized entropy in [0,1]), producing skewed or bimodal distributions rather than normal distributions.
- Heteroscedasticity: Variance in entropy estimates varies systematically across q-values. Low q-values (emphasizing rare transcripts) produce volatile entropy estimates with high variance; high q-values (emphasizing abundant transcripts) produce stable estimates with lower variance.
- Outlier sensitivity: Rare isoforms and count variability can produce extreme entropy values that heavily influence parametric tests.

ART mitigates these issues through alignment and ranking — stripping main effects before ranking preserves interaction structure while removing distributional shape dependence. The aligned ranks are more uniformly distributed and contain no outliers, making the method substantially less sensitive to distributional shape than parametric tests (Higgins and Tashtoush 1994; Wobbrock et al. 2011). Critically, rank ordering is unaffected by normalization choice; whether entropy is normalized, log-transformed, or left raw, the test produces identical results (Conover and Iman 1981). This robustness does not make ART assumption-free: its validity still depends on the experimental design, exchangeability structure, alignment procedure, and sample size.

2. Paired Design for Ordered q-Value Structure

Tsallis entropy has a fundamental sequential property: entropy curves are smooth functions of q . As q increases from 0 (rare-species-emphasizing) to ∞ (common-species-emphasizing), entropy values change systematically. This ordered structure is critical to interpreting q -dependent patterns and exhibits **AR(1) autocorrelation**: consecutive q -values produce correlated entropy estimates.

ART handles this structure through its alignment step: within-subject ranking after stripping main effects preserves the pairing structure while removing the influence of AR(1) dependence (Higgins and Tashtoush 1994; Wobbrock et al. 2011). Because aligned ranks are constructed from residuals after removing main effects, autocorrelation in the original entropy values does not propagate into the test statistic — rank-based tests are invariant to monotonic transformations of the data, and alignment renders the residuals approximately exchangeable under the null hypothesis of no interaction (Ernst 2004; Song 2007). However, this robustness is not absolute: strong autocorrelation can reduce effective sample size and affect power even when the test remains valid.

3. Interaction Testing: Testing $q \times \text{Condition}$ Effects

The implementation uses ART (alignment + ranking + ANOVA), a rank-based factorial approach:

$$F_{q \times \text{condition}} = \frac{MS_{\text{interaction}}}{MS_{\text{residual}}}$$

where:

- Main effects are stripped via alignment: residuals = original - main effect estimates
- Aligned data are then ranked: $R = \text{rank}(\text{aligned residuals})$ (within-subject for paired designs)
- Two-way ANOVA decomposes aligned ranked data: $R = \mu + \alpha_q + \beta_{\text{condition}} + \gamma_{q \times \text{condition}} + \epsilon$
- $MS_{\text{interaction}}$ = sum of squares for $q \times \text{condition}$ interaction / degrees of freedom
- MS_{residual} = residual sum of squares / residual degrees of freedom
- F -statistic follows F -distribution under null hypothesis of no interaction

This directly tests the core biological question: “Does entropy q -dependence differ between groups?” ART is validated across balanced and unbalanced designs (Wobbrock et al. 2011); our Monte Carlo results confirm calibration for the paired design (and flag unpaired ART as mildly anti-conservative at small n).

Assumption Validation

To ensure valid statistical inference, we verify key data assumptions across multiple dimensions.

```
# Validate statistical assumptions (including GAMM diagnostics)
analysis <- calculate_assumptions(
  analysis,
  checks = "all" # Include core checks + GAMM diagnostics
)

# Get assumptions
assumptions_text <- results(analysis, type = "assumptions")
print(assumptions_text)
```

Table 1: **Supplementary Table 14 | Assumption Checks for Aligned Rank Transform (ART)**. Evaluates homogeneity of variance and normality for rank-based interaction test validity.

Test	Result	Interpretation
Exchangeability (Permutation test)	p=0e+00	Ordering detected
Monotonicity (Spearman rho)	r=0.326	Homogeneous
Consistency (Kendall's W / ICC)	W=0e+00, ICC=0.047	Low
Concurvity (Smooth collinearity)	0.000	Low
EDF Ratio (Smoothing)	0.024	Over-smoothed
Non-linearity (Delta R ² vs LM)	0.0%	Use linear
Basis Dimension (Spline basis)	k=3	Adequate
Correlation fit	Observed autocorr=0.076; independence suitable	Good fit
Cluster variation	Mean size=77.0	Homogeneous
Independence	Mean within-cluster residual correlation=0.069	Independent
Scale parameter	phi=0.100	Under-dispersed (rare)
Variance components	ICC=0.049; B=2.14e-04, W=0.004	Use sait model
Normality	p=8.76e-09	Non-normal
Homogeneity	p=2.23e-159; CV=0.872	Heterogeneous
Influence	Outliers=1144, Extreme=894, Influential=8.6%	Some outliers
Variance adequacy	Components for 90%=5, 95%=7, 99%=11	Moderate reduction
Bootstrap stability	Bootstrap SE=0.380; Stable CIs=100%	Moderate stability

Interpretation of Results: The exchangeability test detects strong serial correlation ($p \approx 0$), indicating that consecutive samples are more correlated than expected by chance. This violation is **not problematic** for the Aligned Rank Transform because ART's alignment step removes main effects before ranking, after which the aligned residuals are approximately exchangeable under the null hypothesis of no interaction (Conover and Iman 1981; Higgins and Tashtoush 1994). While strong autocorrelation can reduce effective sample size and affect power, the test itself remains valid.

Running the ART Analysis

We run ART (the default rank method) with Hochberg step-up correction for multiple testing across genes, then compare the results against GAMM below.

```
# Run Aligned Rank Transform (ART, default method)
# This preserves the GAMM results in 'analysis' for later concordance comparison

# Run ART for q × condition interaction
analysis_rank <- calculate_rank_transform(
  analysis,
  multicorr = "hochberg"
)

# Extract ALL results (no filtering) for summary statistics
rank_transform_results_all <- results(analysis_rank, type = "rank_test", rankBy = "pvalue")
```

```
# Extract top significant genes for display
rank_transform_results <- results(analysis_rank, type = "rank_test", rankBy = "pvalue", n = 20, filterF
print(head(rank_transform_results, n = 10))
```

Table 2: **Supplementary Table 15 | Aligned Rank Transform (ART) Results Summary.** Non-parametric test of $q \times$ condition interactions via ART. Compare with GAMM results (main vignette Table 3) to assess parametric vs. non-parametric method concordance.

Metric	Value
Genes tested	77
Significant ($p < 0.05$)	52
Significant ($\text{adj}_p < 0.05$, FWER-controlled)	50
NAs	0
Mean effect size (η^2)	1.6%
Median effect size (η^2)	0.8%
Strong effect genes ($\eta^2 > 10\%$)	1

Table 3: **Supplementary Table 16 | Top genes identified by Aligned Rank Transform (ART).** Ranked by adjusted p-value; comparison with GAMM (main vignette Table 3) reveals method-specific sensitivities.

Gene	P-value	Adj. P-value	F-Statistic	Effect Size (eta-squared)	Interaction Class
CXCL12	3.8e-146	2.9e-144	106.0585	0.1062	Strongly q-dependent
SPNS2	1.1e-111	8.3e-110	56.7003	0.0774	Moderately q-dependent
CCNE1	1.1e-87	8e-86	35.5687	0.0367	Moderately q-dependent
THOC6	1.7e-79	1.3e-77	30.0465	0.0422	Moderately q-dependent
DUOXA2	3e-78	2.2e-76	29.2740	0.0624	Moderately q-dependent
ETFRF1	1.5e-77	1.1e-75	28.8350	0.0227	Moderately q-dependent
LINC03040	8.1e-74	5.8e-72	26.6334	0.0511	Moderately q-dependent
PLP1	7.8e-68	5.5e-66	23.3649	0.0551	Moderately q-dependent
SRPRA	1.5e-58	1.1e-56	18.8734	0.0141	Moderately q-dependent
GLB1L2	2.7e-56	1.8e-54	17.8874	0.0357	Moderately q-dependent

Visualize q-curves for top genes:

```
# Plot top genes from Aligned Rank Transform (ART) (using all genes, not just significant ones)
# This ensures the plot displays n_top genes regardless of significance threshold
top_genes_plot <- plot_diversity_spectrum(analysis, sait_res = rank_transform_results_all, n_top = 4)
print(top_genes_plot)
```

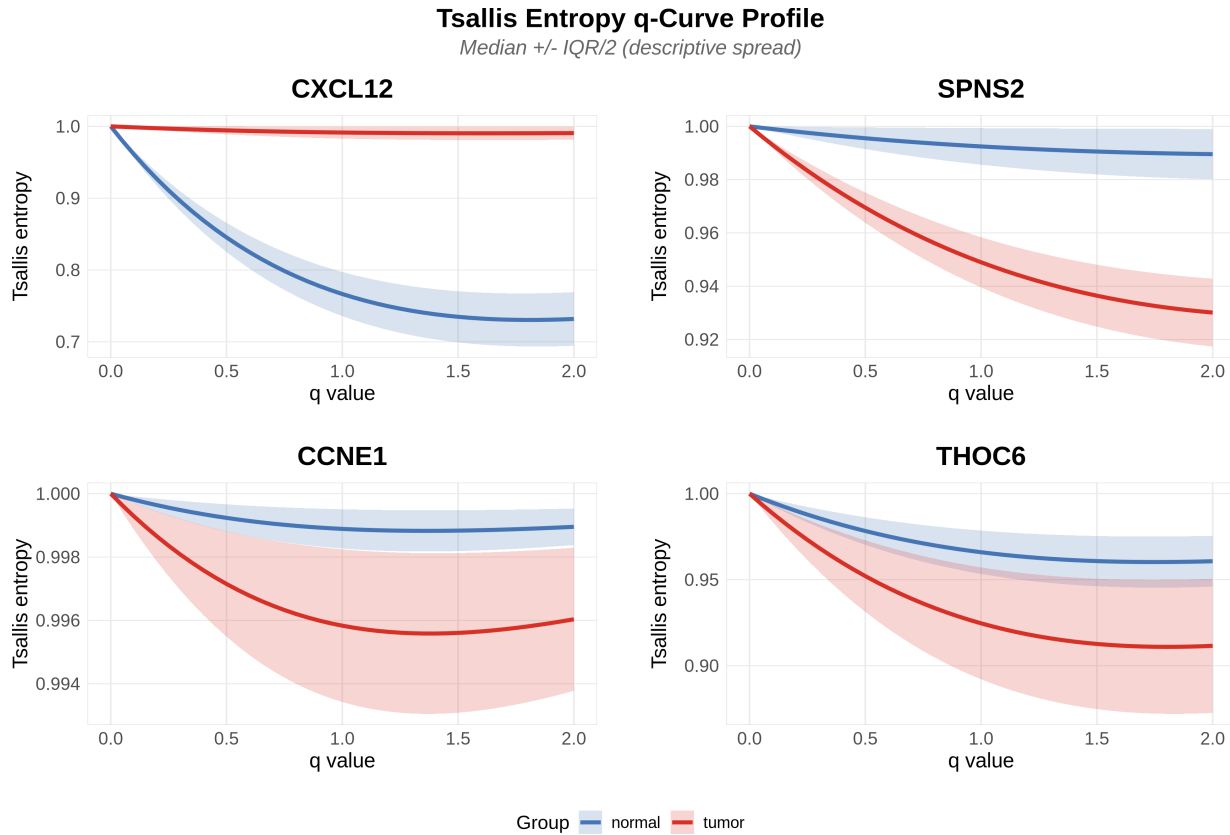


Figure 1: ****Supplementary Figure 1 | Aligned Rank Transform (ART) results for scale-dependent isoform switching.**** Q-curves for top genes identified by ART; compare visual patterns with GAMM results (main vignette Figure 2) to assess method agreement.

Generalized Additive Mixed Model Approach (GAMM)

This section loads precomputed GAMM results generated in the main vignette (TSENAT.Rmd).

```
# Load precomputed GAMM analysis object from RDS file
analysis_sait <- readRDS(
  system.file("extdata", "analysis_sait.rds", package = "TSENAT")
)

# Extract GAMM results for inspection using accessor function
sait_results <- results(analysis_sait, type = "sait", rankBy = "pvalue")
print(sait_results)
```


Table 4: **Supplementary Table 17 | GAMM Results Summary** (summary of the GAMM results computed in the main vignette, Table 3). Overview of generalized additive mixed model statistics for $q \times$ condition interaction tests across full q -spectrum with paired sample structure.

Metric	Value
Genes tested	76
Significant ($p < 0.05$)	66
Concordant ($p < 0.05$ AND $\text{adj_}p < 0.05$)	57
NAs	0
Mean effect size	36.3%
Median effect size	33.3%
Strong effect genes ($\text{effect_size} > 30\%$)	39
Model convergence rate	100.0%

Table 5: **Supplementary Table 18 | Top 10 Genes by GAMM p-value (with Effect Size and Test Statistics)**. Ranked by adjusted p-value; compare with ART results (Supplementary Table 16) for method agreement assessment.

Gene	p (interaction)	Adjusted p	Effect size	Test statistic
CXCL12	2.15e-139	1.63e-137	74.0%	370.00
ING3	4.15e-57	3.12e-55	32.3%	109.18
THY1	7.49e-55	5.54e-53	35.8%	103.97
LINC03040	5.54e-47	4.05e-45	58.3%	86.46
SNHG10	7.69e-43	5.54e-41	21.0%	77.62
MEF2A	2.34e-36	1.66e-34	17.7%	64.30
ASMTL	2.18e-33	1.52e-31	39.4%	58.41
HDAC2	2.93e-33	2.02e-31	22.7%	58.15
ENSG00000274322	3.02e-29	2.05e-27	43.9%	50.38
ATG5	2.32e-28	1.56e-26	51.5%	48.70

Method Concordance Analysis

A critical validation step is to compare whether the Aligned Rank Transform (ART) and GAMM produce consistent results. High concordance between methods (i.e., genes significant in both approaches) provides strong evidence that discoveries are robust to methodological choice. Discordant genes — those significant in only one method — warrant closer inspection: they may represent genuine biological signals revealed by one method’s specific advantages, or artifacts of that method’s assumptions. This section quantifies agreement between approaches and identifies high-confidence genes significant across both statistical frameworks.

```
# Compute concordance analysis using new two-object API
rank_results <- results(analysis_rank, type = "rank_test")
if (is.null(rank_results) || length(rank_results) == 0) {
  stop("Cannot run calculate_concordance(): rank_test results are empty. Ensure calculate_rank_transf")
}
```

```
analysis_rank <- calculate_concordance(
  analysis_sait = analysis_sait,
  analysis_rank = analysis_rank,
  verbose = TRUE
)

# Extract concordance results with list format for component display
concordance_results <- results(analysis_rank, type = "concordance", format = "list")
```

Table 6: **Supplementary Table 19 | Global Concordance Metrics: GAMM vs Aligned Rank Transform (ART).** Quantifies agreement between parametric (GAMM) and non-parametric (ART) approaches for detecting $q \times$ condition interactions.

Metric	Value
Total genes compared	76
Spearman correlation (p-values)	rho = 0.3082
Both methods significant ($p < 0.05$)	40 (52.6%)
SAIT only significant	17 (22.4%)
Rank test only significant	9 (11.8%)
Neither significant	10 (13.2%)
Concordance rate	52.6%
Discordance rate	34.2%

Table 7: **Supplementary Table 20 | Robust Entropic Order Index Interactions: Top 10 High-Confidence Genes Detected by Both Methods (of 40 concordant, ranked by statistical significance).** Cross-method validation: genes significant in both GAMM and ART provide strongest evidence for $q \times$ condition effects.

Gene	SAIT adj p	Rank test adj p	SAIT Effect	Rank test rho ²
CXCL12	1.63e-137	2.95e-144	74.0%	0.106
LINC03040	4.046e-45	5.79e-72	58.3%	0.051
SNHG10	5.538e-41	5.655e-49	21.0%	0.030
MEF2A	1.662e-34	7.61e-53	17.7%	0.043
FAXDC2	1.438e-21	1.288e-32	19.4%	0.033
ING3	3.12e-55	6.756e-21	32.3%	0.014
FAM114A2	6.857e-19	2.73e-54	39.5%	0.037
ETFRF1	1.935e-18	1.11e-75	62.5%	0.023
ATG5	1.557e-26	3.117e-18	51.5%	0.013
SPNS2	4.395e-18	8.25e-110	4.4%	0.077

Statistical Power vs. Robustness: Understanding Method Discordance

The substantial discordance between GAMM and ART results — where GAMM identifies many more significant genes — reflects a fundamental trade-off in statistical methodology: **parametric methods maximize power when assumptions hold, while non-parametric methods sacrifice power for robustness**

to assumption violations. The exact percentages depend on the dataset and analysis parameters; the pattern of GAMM detecting more interactions than ART is the consistent finding.

GAMM's superior power derives from three factors:

1. **Distributional assumptions:** GAMM assumes approximately normal residuals and homogeneous variance, which are reasonable after appropriate transformation for entropy data. Under these assumptions, parametric methods are theoretically optimal, achieving maximum power for a given type I error rate. ART is assumption-free but necessarily discards quantitative information through alignment and ranking, which reduces statistical power when the underlying data are approximately normal.
2. **Model flexibility:** GAMM captures nonlinear q patterns with splines (natural regression splines with 3 df for paired designs) that simultaneously fit nonlinear trends while controlling degrees of freedom. This flexibility allows GAMM to detect subtle entropic index $\times$ group interactions across all q -values.
3. **Continuous vs. categorical q treatment:** GAMM models q as a continuous smooth predictor across all 41 q -values, preserving the ordinal structure and borrowing strength across adjacent q -levels. ART treats q as a categorical factor — while this enables proper interaction testing via alignment, it discards the continuous ordering information, reducing sensitivity to graded q -dependent trends. This discretization is a major contributor to ART's lower power relative to GAMM, independent of distributional assumptions.

This complementary approach—combining high-power parametric tests with robust nonparametric alternatives—provides confidence that discoveries are methodologically sound rather than artifacts of statistical assumptions.

Visualization: Method Comparison

Visual comparison of concordance patterns provides intuitive assessment of which genes align between methods and which are method-specific. The following plots display significance calls, effect sizes, and agreement metrics in a format that facilitates interpretation of both robust discoveries and method-specific findings.

```
# Create comparison plot using S4 wrapper  
# Concordance analysis results are stored in analysis metadata after calculate_concordance()  
p <- plot_concordance(analysis_rank, verbose = TRUE)  
grid::grid.draw(p)
```

Comparing interaction detection across two statistical methods

Concordance analysis between SAIT and ART/Conover-Iman Rank Transform tests

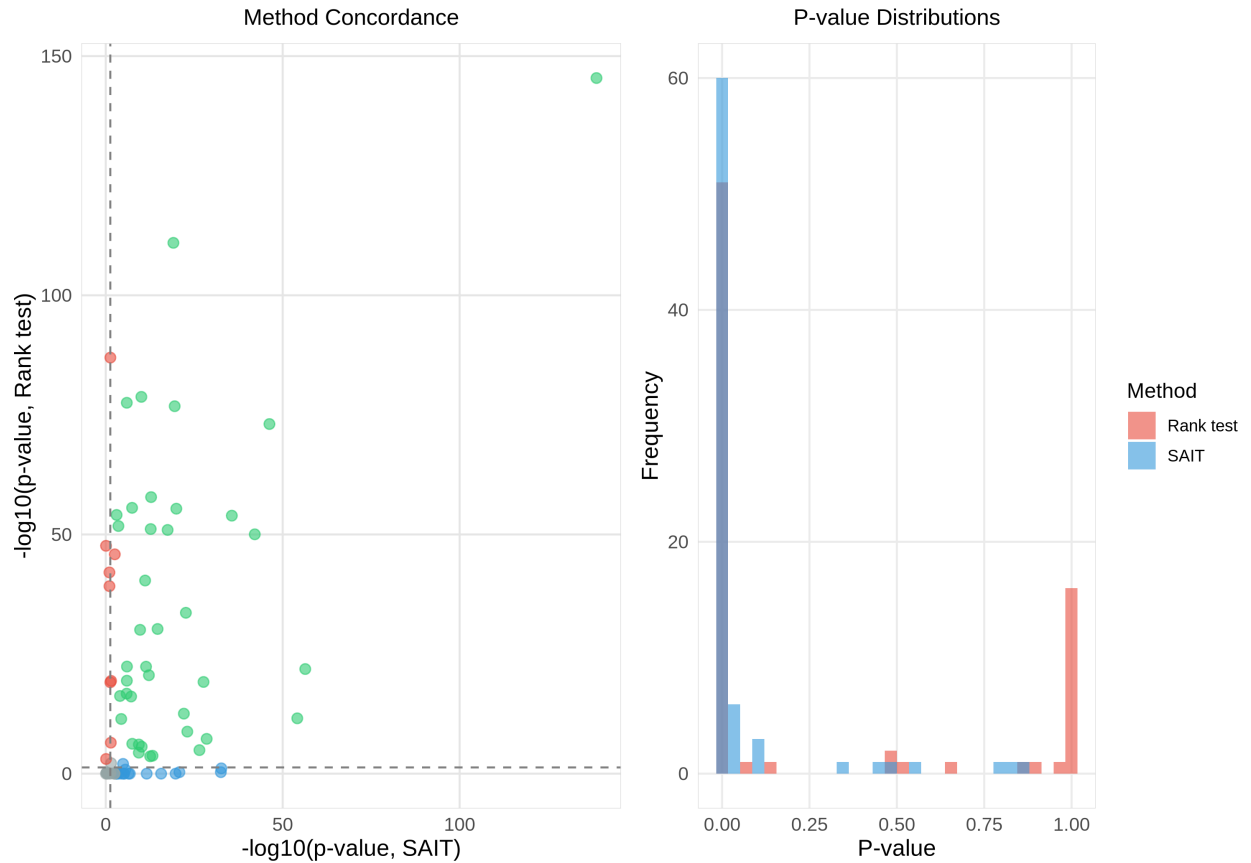


Figure 2: **Supplementary Figure 2 | Method concordance visualization comparing Aligned Rank Transform (ART) and GAMM approaches for detecting $q \times \text{condition}$ interactions.** Scatter plot showing overlap in genes identified as significant by each method; reveals parametric-non-parametric agreement patterns.

Session Information

```
sessionInfo()
#> R version 4.5.3 (2026-03-11)
#> Platform: x86_64-conda-linux-gnu
#> Running under: Ubuntu 22.04.5 LTS
#>
#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblas-p-r0.3.30.so; LAPACK version 3.12.0
#>
#> locale:
#> [1] LC_CTYPE=es_ES.UTF-8      LC_NUMERIC=C
#> [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=es_ES.UTF-8
```

```

#> [5] LC_MONETARY=de_DE.UTF-8    LC_MESSAGES=es_ES.UTF-8
#> [7] LC_PAPER=de_DE.UTF-8      LC_NAME=C
#> [9] LC_ADDRESS=C              LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats4      stats      graphics  grDevices  utils      datasets  methods
#> [8] base
#>
#> other attached packages:
#> [1] TSENAT_0.99.35           gridExtra_2.3
#> [3] ggplot2_4.0.3            knitr_1.51
#> [5] dplyr_1.2.1              SummarizedExperiment_1.40.0
#> [7] Biobase_2.70.0           GenomicRanges_1.62.1
#> [9] Seqinfo_1.0.0            IRanges_2.44.0
#> [11] S4Vectors_0.48.1         BiocGenerics_0.56.0
#> [13] generics_0.1.4          MatrixGenerics_1.22.0
#> [15] matrixStats_1.5.0        kableExtra_1.4.0
#> [17] BiocStyle_2.38.0         testthat_3.3.2
#>
#> loaded via a namespace (and not attached):
#> [1] Rdpack_2.6.6             sandwich_3.1-1           rlang_1.2.0
#> [4] magrittr_2.0.5           multcomp_1.4-30         otel_0.2.0
#> [7] compiler_4.5.3           mgcv_1.9-4              systemfonts_1.3.2
#> [10] vctrs_0.7.3              stringr_1.6.0           pkgconfig_2.0.3
#> [13] fastmap_1.2.0            magick_2.9.1            ellipsis_0.3.3
#> [16] backports_1.5.1          XVector_0.50.0          labeling_0.4.3
#> [19] rmarkdown_2.31           sessioninfo_1.2.3       tzdb_0.5.0
#> [22] nloptr_2.2.1             tinytex_0.59            purrr_1.2.2
#> [25] xfun_0.58                cachem_1.1.0            jsonlite_2.0.0
#> [28] DelayedArray_0.36.1      BiocParallel_1.44.0     broom_1.0.13
#> [31] parallel_4.5.3           R6_2.6.1                stringi_1.8.7
#> [34] bslib_0.11.0             RColorBrewer_1.1-3      car_3.1-5
#> [37] boot_1.3-32              pkgload_1.5.2           brio_1.1.5
#> [40] jquerylib_0.1.4          estimability_1.5.1      bookdown_0.46
#> [43] Rcpp_1.1.1-1.1          usethis_3.2.1           zoo_1.8-15
#> [46] readr_2.2.0              Matrix_1.7-5            splines_4.5.3
#> [49] tidyselect_1.2.1         rstudioapi_0.18.0       abind_1.4-8
#> [52] yaml_2.3.12              codetools_0.2-20        pkgbuild_1.4.8
#> [55] lattice_0.22-9           tibble_3.3.1            plyr_1.8.9
#> [58] withr_3.0.3              S7_0.2.2               evaluate_1.0.5
#> [61] desc_1.4.3               survival_3.8-6          xml2_1.5.2
#> [64] BiocManager_1.30.27      pillar_1.11.1           carData_3.0-6
#> [67] reformulas_0.4.4         rprojroot_2.1.1         hms_1.1.4
#> [70] scales_1.4.0             minqa_1.2.8             ARTool_0.11.2
#> [73] xtable_1.8-8             glue_1.8.1              pheatmap_1.0.13
#> [76] emmeans_2.0.3           tools_4.5.3             lme4_2.0-1
#> [79] fs_2.1.0                 mvtnorm_1.3-6           cowplot_1.2.0
#> [82] grid_4.5.3              tidyr_1.3.2             rbibutils_2.4.1
#> [85] devtools_2.5.2          nlme_3.1-169            Formula_1.2-5

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#> [88] cli_3.6.6          textshaping_1.0.5  viridisLite_0.4.3
#> [91] S4Arrays_1.10.1    suglite_2.2.2      geepack_1.3.13
#> [94] gtable_0.3.6       sass_0.4.10        digest_0.6.39
#> [97] SparseArray_1.10.10 TH.data_1.1-5      farver_2.1.2
#> [100] memoise_2.0.1      htmltools_0.5.9    lifecycle_1.0.5
#> [103] MASS_7.3-65
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